

Dickman Approximation using Stein's Method

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Perpetuity

Suppose at time $t = 0$, you have an asset worth W_0 units.

Let $n \geq 0$. At the n^{th} time point, a deposit scaled to be unity is added to the asset.

During that period, the value of the asset depreciates by an independent Uniform factor.

$$W_{n+1} =_d U_{n+1}(W_n + 1).$$

What is its stationary distribution?

Dickman Distribution

Fix $\theta > 0$. For non-negative W and $U \sim \mathbf{U}[0, 1]$ independent of W , consider its θ -Dickman bias transformation

$$W^* =_d U^{1/\theta}(W + 1).$$

The generalized Dickman distribution $\mathcal{D}_\theta$ can be defined as its unique fixed point.

We will only consider the case $\theta = 1$ here. Denote a standard Dickman random variable by D with distribution $\mathcal{D}$.

Dickman Distribution

First appeared in Dickman'30 in the context of *smooth numbers*.

For x and y positive, let

$$\psi(x, y) = \# \text{ of integers } \leq x \text{ with all its prime factors } \leq y.$$

Then for all $u > 0$,

$$\psi(x, x^{1/u}) \sim x\rho(u) \text{ as } x \rightarrow \infty$$

where ρ is the Dickman density function up to a scaling.

Since then, have appeared in logarithmic combinatorial structures, algorithms, probabilistic number theory etc.

Dickman Convergence: Earlier Works

- Runtime of Quickselect sorting algorithm to find the m^{th} smallest element from a list of n distinct numbers.
 - Hwang and Tsai'02 showed weak convergence.
 - A $O(\log n/n)$ rate of convergence was proved in Goldstein'17.
- Sum of Records: For $\sigma \in \mathcal{S}_n$ a random permutation, let S_n be the sum of positions of records which are positions k such that $\sigma(k) > \sigma(1), \dots, \sigma(k-1)$.
 - One can show that $S_n = \sum_{k=1}^n k B_k$ with $B_k \sim \text{Ber}(1/k)$ independent. Moreover, $S_n/n \xrightarrow{d} D$.

- Probabilistic number theory: Let $p_n, n \geq 1$ be an enumeration of primes in increasing order.
 - Let Ω_n be the set of integers with largest possible prime factor p_n . Consider a random variable $M_n \in \Omega_n$ with $P(M_n = x) \propto 1/x$.
 - Pinsky'16 showed

$$(\log p_n)^{-1} \log M_n \xrightarrow{d} D.$$

- Same holds true when Ω_n is replaced by Ω'_n , the set of square-free integers with largest possible prime factor p_n .

Goal is to study Dickman convergence and get bounds using Stein's method.

Outline

- 1 Stein's Method
- 2 Dickman Approximation
- 3 Application in Summations

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The Central Limit Theorem

Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2 . Then for Z a standard normal random variable,

$$W_n = \frac{S_n - n\mu}{\sqrt{n}\sigma} \xrightarrow{d} Z \quad \text{as } n \rightarrow \infty.$$

If in addition, $\mathbb{E}|X_1|^3 < \infty$, a bound on the distance between the distribution function F_n of W_n and Φ is provided by the Berry-Esseen Theorem,

$$\sup_{x \in \mathbb{R}} |F_n(x) - \Phi(x)| \leq \frac{c\mathbb{E}|X_1 - \mu|^3}{\sqrt{n}\sigma^3}.$$

Distance

Let X and Y be random variables with distributions μ and ν . One way to measure the distance between μ and ν is to define

$$d_{\mathcal{H}}(X, Y) = \sup_{h \in \mathcal{H}} |\mathbb{E}h(X) - \mathbb{E}h(Y)|$$

for some class $\mathcal{H}$, the class of test functions.

- Kolmogorov metric: For $\mathcal{H} = \{\mathbf{1}(-\infty, x] : x \in \mathbb{R}\}$,

$$d_{\mathcal{K}}(X, Y) = \sup_{x \in \mathbb{R}} |\mu(-\infty, x] - \nu(-\infty, x]|.$$

- Total variation: For $\mathcal{H} = \{\mathbf{1}(B) : B \in \mathcal{B}\}$,

$$d_{TV}(X, Y) = \sup_{B \in \mathcal{B}} |\mu(B) - \nu(B)|.$$

Wasserstein-1, or L^1 distance

Taking $\mathcal{H}$ as

$$\mathcal{H} = \text{Lip}_1 = \{h : |h(x) - h(y)| \leq |x - y|\},$$

one obtains the Wasserstein-1 metric

$$d_1(X, Y) = \sup_{h \in \text{Lip}_1} |\mathbb{E}h(X) - \mathbb{E}h(Y)|.$$

One can also write

$$d_1(X, Y) = \inf \mathbb{E}|X - Y|$$

over all couplings of X and Y .

To bound $d_1(X, Y)$, it is enough to find a nice coupling.

Stein's Lemma

Characterization of the $\mathcal{N}(0, 1)$ distribution: $Z \sim \mathcal{N}(0, 1)$ if and only if

$$\mathbb{E}[Zf(Z)] = \mathbb{E}[f'(Z)]$$

for all absolutely continuous functions f with $\mathbb{E}|f'(Z)| < \infty$.

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Basic idea: For a random variable W and a sufficiently large class of functions $\mathcal{F}$, if

$$\mathbb{E}Wf(W) \approx \mathbb{E}f'(W) \quad \text{for all } f \in \mathcal{F},$$

then expect $W \approx \mathcal{N}(0, 1)$.

From Stein's Lemma to Stein Equation

Stein's Lemma says the discrepancy between W and Z is also reflected in

$$\mathbb{E}[f'(W) - Wf(W)].$$

Can measure distance from W to a standard normal Z by

$$\mathbb{E}h(W) - Nh$$

where $Nh = \mathbb{E}h(Z)$.

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Equating the two, we obtain the Stein equation:

$$f'(w) - wf(w) = h(w) - Nh.$$

Stein Equation

$$f'(w) - wf(w) = h(w) - Nh. \quad (\star)$$

Hence

$$\mathbb{E}[f'(W) - Wf(W)] = \mathbb{E}h(W) - Nh.$$

Recall, for $h \in \mathcal{H}$, we want to compute (a bound on)

$$\mathbb{E}h(W) - Nh.$$

Approach: Given h , solve $(\star)$ for f , and instead compute

$$\mathbb{E}[f'(W) - Wf(W)].$$

Solution and Smoothness

Given $h \in \mathcal{H}$,

$$f_h(w) = e^{w^2/2} \int_{-\infty}^w (h(z) - Nh) e^{-z^2/2} dz$$

is the unique bounded solution to $(\star)$. Moreover for $h \in \text{Lip}_1$,

$$\|f_h\|_{\infty} \leq 2, \|f'_h\|_{\infty} \leq \sqrt{2/\pi}, \|f''_h\|_{\infty} \leq 2.$$

Need to compute

$$\mathbb{E}[f'(W) - Wf(W)]$$

which typically involves introducing an auxiliary random variable coupled to W having certain properties. [Size biasing](#), [Zero biasing](#), [Exchangeable pairs](#) ...

Zero Bias Transformation

Generally, distributional characterizations provide an additional avenue to study approximations.

For W with mean zero and variance one, say W^z has W -zero biased distribution if

$$\mathbb{E}Wf(W) = \mathbb{E}f'(W^z)$$

for all absolutely continuous f .

$\mathcal{N}(0, 1)$ is the unique fixed point of this transformation.

Closeness of W^z to W gives an idea about closeness of W to $Z \sim \mathcal{N}(0, 1)$:

$$d_1(W, Z) \leq 2d_1(W, W^z).$$

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Compare to Zero Bias Transform

Looking at the associated Markov chain, one can show that $W \sim \mathcal{D}$ iff for $g \in \text{Lip}_1$,

$$\mathbb{E}A_{W+1}g = \mathbb{E}g(W) \quad \text{where } A_xg = \frac{1}{x} \int_0^x g(u)du \text{ for } x > 0.$$

Generally,

$$\mathbb{E}A_{W+1}g = \mathbb{E}g(W^*) \quad \text{where } W^* =_d U(W + 1).$$

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Generally,

$$\mathbb{E}A_{W+1}g = \mathbb{E}g(W^*) \quad \text{where } W^* =_d U(W+1).$$

For $X \sim F$ with zero mean variance one, $\mathcal{N}(0, 1)$ is the unique fixed point of the zero bias transform

$$\mathbb{E}Xf(X) = \mathbb{E}f'(X^z).$$

Also,

$$d_1(X, Z) \leq 2d_1(X^z, X).$$

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Generally,

$$\mathbb{E}A_{W+1}g = \mathbb{E}g(W^*) \quad \text{where } W^* =_d U(W+1).$$

For W non-negative, $\mathcal{D}$ is the unique fixed point of the Dickman bias transform

$$\mathbb{E}A_{W+1}g = \mathbb{E}g(W^*).$$

Goldstein'17 showed

$$d_1(W, D) \leq 2d_1(W^*, W).$$

Alternative Stein Equation

Generally, not easy to construct a nice coupling between W and W^* and appears elusive for most examples.

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Alternative characterization: A non-negative random variable $W \sim \mathcal{D}$ iff

$$\mathbb{E}[W f'(W) + f(W) - f(W + 1)] = 0 \quad \text{for all } f \in \cup_{\alpha \geq 0} \text{Lip}_{\alpha}.$$

Stein equation: We consider the Stein equation

$$x f'(x) + f(x) - f(x + 1) = h(x) - \mathbb{E}[h(D)].$$

Solution and Smoothness Bounds

Stein equation:

$$xf'(x) + f(x) - f(x+1) = h(x) - \mathbb{E}[h(D)]. \quad (\star\star)$$

Define the class of twice differentiable functions

$$\mathcal{H}_{\alpha,\beta} = \{h : h \in \text{Lip}_{\alpha}, h' \in \text{Lip}_{\beta}\} \quad \text{for } \alpha \geq 0, \beta \geq 0.$$

Smoothness Bound (B., Goldstein)

For $h \in \mathcal{H}_{1,1}$, there exists a solution $f \in \mathcal{H}_{1,1/2}$ to $(\star\star)$.

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Weighted Bernoulli Sums

Consider $W_n = n^{-1} \sum_{k=1}^n kB_k$, $B_k \sim \text{Ber}(1/k)$ independent.

$$\begin{aligned}\mathbb{E}h(W_n) - \mathbb{E}[h(D)] &= \mathbb{E}[W_n f'(W_n)] - \mathbb{E}[f(W_n + 1) - f(W_n)] \\ &= \mathbb{E}[W_n f'(W_n)] - \mathbb{E}[f'(W_n + U)]\end{aligned}$$

with $U \sim \mathbf{U}[0, 1]$ independent of W_n .

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For D a standard Dickman random variable, from the characterization one has

$$\mathbb{E}[D\phi(D)] = \mathbb{E}[\phi(D + U)]$$

where $U \sim \mathbf{U}[0, 1]$ independent of D .

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$$\mathbb{E}[D\phi(D)] = \mathbb{E}[\phi(D + U)]$$

where $U \sim \mathbf{U}[0, 1]$ independent of D .

With $\bar{U} \sim \mathbf{U}\{1/n, \dots, k/n, \dots, 1\}$ independent of W_n ,

$$\mathbb{E}[W_n f'(W_n)] \approx \mathbb{E}[f'(W_n + \bar{U})].$$

An Improved Bound

Thus,

$$\mathbb{E}h(W_n) - \mathbb{E}[h(D)] \approx \mathbb{E}[f'(W_n + \bar{U})] - \mathbb{E}[f'(W_n + U)].$$

Define the Wasserstein-2 metric as

$$d_{1,1}(X, Y) = \sup_{h \in \mathcal{H}_{1,1}} |\mathbb{E}h(Y) - \mathbb{E}h(X)|.$$

Optimal Bound: $d_{1,1}(W_n, D) \leq (3/4)n^{-1}.$

We improve on the rate $O(\sqrt{\log n}/n)$ in Arras et. al.'17 and provide bounds in stronger Wasserstein-2 metric, also in Kolmogorov metric.

For D a standard Dickman random variable, it satisfies,

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Heuristically, for S with mean 1, if one can find T possibly weakly dependent on S such that

$$\mathbb{E}[S\phi(S)] \approx \mathbb{E}[\phi(S + T)],$$

then closeness of T to $U \sim \mathbf{U}[0, 1]$ independent of S gives an idea about closeness of S and D .

Size Biasing Contd.

Size Bias Theorem (B., Goldstein)

Let S be non-negative with mean μ and finite variance such that for some random variable T ,

$$\mathbb{E}[S\phi(S)] = \mu\mathbb{E}[\phi(S+T)] + R_\phi \quad \text{for all } \phi \in \text{Lip}_{1/2}.$$

Then

$$d_{1,1}(S, D) \leq |\mu - 1| + \frac{1}{2} \inf_{(T,U)} \mathbb{E}|T - U| + \sup_{\phi \in \text{Lip}_{1/2}} |R_\phi|$$

where the infimum is over all couplings (T, U) constructed on the same space as S , with U independent of S .

We say $S + T$ is an approximate size biased version of S .

Reciprocal Distribution on Ω_n

For $p_n, n \geq 1$ an enumeration of primes in increasing order and Ω_n the set of integers with largest possible prime factor p_n , let $M_n \sim \Pi_n$ where

$$\Pi_n(x) \propto \frac{1}{x} \quad \text{for } x \in \Omega_n.$$

Writing $M_n = \prod_{k=1}^n p_k^{X_k}$, It can be shown that $X_k \sim \text{Geom}(1 - 1/p_k)$ and are independent.

$$\frac{\log M_n}{\log(p_n)} =_d \frac{1}{\log(p_n)} \sum_{k=1}^n X_k \log(p_k) =: S_n.$$

To prove bounds, need approximate size biasing for S_n .

Approximate Size Biasing

Pick I independent proportional to the mean of summands:

$$P(I = k) = \frac{\mathbb{E}(X_k) \log(p_k)}{\mu_n \log(p_n)} \quad \text{for } k \in \{1, \dots, n\}.$$

With $T_n = \log(p_I)/\log(p_n)$, $S_n + T_n$ satisfies

$$\mathbb{E}[S_n \phi(S_n)] \approx \mu_n \mathbb{E}[\phi(S_n + T_n)].$$

By Size Bias Theorem, we obtain

Optimal Bound: $d_{1,1}(S_n, D) = O(1/\log n).$

Square-free Numbers

A closely related problem is to consider Ω'_n , the set of square-free integers with largest possible prime factor p_n and let Π'_n denote the reciprocal distribution on Ω'_n .

One can show $M_n = \prod_{k=1}^n p_k^{X_k} \sim \Pi'_n$ where $X_k \sim \text{Ber}(1/(1 + p_k))$ are independent.

Arras et. al.'17 proved a $O(\log \log n)^{3/2}(\log n)^{-1}$ rate of convergence for $\log M_n / \log p_n$ in Wasserstein-3 metric.

Using the Size Bias Theorem, we provide an optimal $O(1/\log n)$ rate in stronger Wasserstein-2 metric.

Further Directions

Dickman distribution also appears in the context of *Minimal Directed Spanning Trees*, but the model appears hard to analyze.

One can understand more by exploring connections to scale invariant Poisson processes.

For the number theoretic examples, it might be possible to obtain Wasserstein-1 bound if one can couple $S_n = \log M_n / \log p_n$ to its Dickman bias transform.

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Thank you